ABDELRAHMAN AHMED ABDELAZEEZ

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SUMMARY

Computer Science undergraduate (MSA University, expected 2028) who has designed, documented and built web systems for a real client, a medical-equipment maintenance company in Giza. Works in PHP/MySQL and the MERN stack (including a real-time restaurant POS), with C++ algorithm projects and IoT/machine-learning work from the University of Greenwich. 1st place (of 5 teams) at the MSA Deep Minds Competition; Vice President of a 280+ member competitive programming club.

EDUCATION

BSc, Computer Science, MSA University *Sep 2024 – Jun 2028*

Coursework: Data Structures and Algorithm Analysis, Web Programming, OOP, Discrete Mathematics

SKILLS

Languages: C++, C, C#, JavaScript, PHP, Python, SQL, HTML/CSS

Web: PHP, JavaScript, jQuery, MySQL, Git/GitHub, Postman, React, Node.js, MongoDB

Embedded and data: Arduino, ESP8266, Proteus, Linux/Bash, pandas, NumPy, scikit-learn

Spoken languages: Arabic (Native), English (Proficient)

EXPERIENCE

Freelance Full-Stack Developer, Zahra Medical, Giza *Sep 2025 – Apr 2026*

- Built the front and back end of the company's public website, now its primary online presence (HTML, CSS, JavaScript, PHP, jQuery).
- Built a PHP equipment booking system with JavaScript validation so the team can schedule equipment maintenance in real time, reducing invalid form submissions.
- Wrote a formal SRS for a PHP/MySQL service platform: 65 functional requirements, role-based access for 7 user roles, ticket lifecycle, SLA and audit logging.
- Delivered the platform as a live, responsive multi-page product: maintenance booking, quote requests, spare-parts inventory, SLA proposal contact and an interactive downtime-cost calculator: abdelrahman-ah1.github.io/ZahraMedical

LEADERSHIP

Vice President, MSA Competitive Programming Club (MSA CPC) *Sep 2026 – Present*

- Lead a club of 280+ students and 18 staff, organizing sessions and events.
- Teach algorithm and data-structure workshops twice a week, with 78% average attendance.

PROJECTS

Dinylo POS + Kitchen Display System | React, Node.js, MongoDB, WebSockets | GitHub *Sep 2026*

- Built a full-stack restaurant POS and kitchen system from a formal SRS: multi-tenant data model, table management, order lifecycle (placed, in prep, ready, served) and QR self-ordering for dine-in tables.
- Pushed order updates to the kitchen in under 1 second over a WebSocket hub on the branch LAN, with automatic resync on reconnect and an offline outbox that replays writes after outages.
- Added ESC/POS receipt and ticket printing over TCP, payment idempotency keys, a role-based audit log, recipe-based stock auto-deduction with low-stock alerts, and an end-of-day reconciliation report.

IoT DC Motor Predictive Maintenance | Python, ESP8266 | University of Greenwich (team project) *Aug 2026*

- Built an ESP8266 system measuring motor speed, temperature, humidity and vibration, with a web dashboard for motor control and CSV logging.
- Labelled vibration as high/low with k-means ($k = 2$, silhouette ≈ 0.98), then trained a k-NN classifier using 5-fold cross-validation.
- Compared AR, ARIMA, SARIMA and Holt-Winters forecasts by MAE and RMSE on held-out data, and built regression models scored by MSE.

2D Arcade Game | C#, Windows Forms | GitHub *May 2026*

- Implemented sprite-sheet rendering, character state transitions and axis-aligned collision detection.

Huffman Compression Tool | C++ | GitHub *Jan 2026*

- Built a lossless .bmp compressor and cut frequency-analysis preprocessing from $O(N^2)$ to $O(N)$.
- Packed variable-length bitstreams into bytes with bit-perfect round trips; fixed memory leaks in tree cleanup.

ACHIEVEMENTS

1st Place of 5 teams, MSA Deep Minds Competition (first-year category) *Jan 2025*

- Built an automated smart-home irrigation system on Arduino Uno.

TRAINING

University of Greenwich Summer Abroad, Faculty of Engineering & Science *Aug 2026*

- 7-day program: IoT microcontrollers, Proteus, Linux, networking, cybersecurity basics, time-series ML.